OPENAI SOLUTIONS ENGINEERING

From fragmented supply-chain signals to governed decisions

A proposed OpenAI platform approach for Zeiss Supply Chain Management and IT leadership

TL;DR

OpenAI becomes the governed decision layer between ZEISS supply-chain data and operational action.

- Executive chat experience
- Enterprise integration and control layer

Problem

Context is split across SAP, logistics, suppliers and files.

Platform move

A central communication interface gives leaders **one decision surface**; the API integrates tools and controls.

Pilot

Start with one high-value workflow and validate faster triage, evidence quality and governed approvals.

Status quo and validation needs

Known operating context

- Advanced manufacturing, medical technology, quality and optics domains
- Decisions draw on SAP, supplier, logistics and document context
- Value depends on trustworthy answers, not another dashboard

Discovery hypotheses to validate

- Which disruptions consume the most time?
- Where do handoffs reduce accuracy or predictability?
- Which recommendations need evidence, review and traceability?

BUSINESS PROBLEM

Providing a decision framework for existing data

SAP

- POs
- Material master
- MRP exceptions

Warehouse

- EWM stock
- Goods receipts
- Production buffers

Carriers

- DHL / FedEx
- ETA changes
- Customs events

Suppliers

- Commit dates
- Capacity
- Quality status

Files

- Scorecards
- Excel trackers
- Contracts

PAIN

- Manual effort to reconstruct one trusted view
- Answer accuracy varies by source snapshot
- Low precision for lead-time and supplier risk
- Reliability depends on scarce experts

COST

- Expedite spend and buffers absorb uncertainty
- Leadership time shifts into escalation meetings
- Late mitigation creates schedule churn
- Scattered evidence increases quality exposure

Supply Chain Hub as the enterprise decision interface

Communication interface

An executive-grade conversation surface where teams ask business questions, inspect evidence and **align on next steps**.

Enterprise operating model

Role scope, source ownership, approval paths and auditability turn the app into a **governed way of working**.

System integration layer

Connects ZEISS systems, documents, business tools and workflow actions, constantly **keeping these sources of truth up to date**.

BUSINESS OUTCOME

- Reduce manual effort by consolidating scattered supply-chain context into one decision interface.
- Improve decision precision with source-backed recommendations and visible assumptions.
- Give leaders a reliable path from signal to recommended action and approval.

Solution overview

Supply Chain Hub
Conversation layer for asking, reviewing evidence and aligning next steps.

→

Intelligence layer
Turns supply-chain questions into options and next steps.

→

Operating controls
Keeps roles, sources, masking and review paths clear.

→

ZEISS systems
SAP, logistics, supplier, warehouse and documents stay sources of truth.

Ground
Pull the relevant supply-chain context.

Reason
Compare risk, options and trade-offs.

Act
Send recommendations to owners or approval paths.

OpenAI Responses API
Streams grounded answers with source context and model controls.

OpenAI Agents SDK
Runs tool-backed workflows and reviewer handoffs.

Why OpenAI / ROI
Faster escalation cycles, less expedite exposure and a reusable business case.

WORKFLOW EXAMPLE

Executive workflow: from risk to next step

Trigger
Customer shipment at risk: supplier date, SAP availability and carrier ETA do not match

→

Question
Can we keep the promise date? What needs approval today?

→

Evidence
Orders, inventory, supplier commits, carrier ETA and expedite cost

→

Action
Recommended option, customer impact, owner and approval step

Before
Teams spend the first escalation meeting reconciling SAP exports, supplier emails and freight updates.

After
Leaders see the exposed orders, root cause, options, cost trade-off and approval owner **in one interface**.

What changes in the room
The meeting starts with a shared view and a recommended next step.

Why this matters
ZEISS keeps source ownership intact while leaders move faster.

Demo

Impact validation and success criteria

The first POC must prove both business impact and operating trust: faster decisions, better evidence and controlled action paths.

1

Impact validation

Baseline
Measure current risk-review cycle time, escalation volume and manual triage hours.

Run the workflow
Use one live supply-chain risk scenario with real source boundaries and named review.

Read out value
Compare avoided effort, decision speed and quality of evidence against baseline.

2

Measurable success criteria

25%
Faster weekly risk-review cycle versus baseline.

90%
Answers cite trusted ZEISS sources and show assumptions.

80%
Target users rate outputs useful for real workflows.

100%
High-impact actions route to accountable review.

FUTURE POTENTIAL

Supply Chain Hub as the business blueprint

Procurement

- Supplier negotiations
- Contract Q&A
- Spend-risk trade-offs

Quality

- Deviation triage
- Corrective action summaries
- Supplier quality signals

Manufacturing

- Line impact analysis
- Work instruction support
- Maintenance handoffs

Finance

- Working capital views
- Expedite-cost exposure
- Scenario assumptions

The repeatable blueprint is the same: conversational workspace, integration layer, enterprise context and accountable workflow ownership.

Proof of value including guardrails

Four-week proof of concept maximum, followed by a business decision on pilot scale.

WEEK 0

Align

Sponsor, workflow owner, success metrics, data boundaries.

WEEK 1

Connect

Representative sources, role scopes, approval policy.

WEEK 2

Build

App workflow, business tools, grounding and audit trail.

WEEK 3

Validate

User feedback, answer quality checks, security review.

WEEK 4

Decide

Impact readout, guardrail review, pilot recommendation.

What is needed to get started: executive sponsorship, access to representative systems and documents, a named business process owner and agreement on operating boundaries for the first controlled pilot.

Annex

Behind the scenes - actual Supply Chain Hub stack

App experience

- Next.js App Router, React and strict TypeScript
- `app/supply-chain-app.tsx` uses AI SDK `useChat`
- Persona, model, thinking and source-selection controls

OpenAI chat path

- `/api/chat` validates requests and builds server context
- `@ai-sdk/openai` calls `openai.responses( ... )`
- Vercel AI SDK streams output; deterministic demo stream remains fallback

Agentic action path

- `/api/actions` checks workflow and persona access
- `@openai/agents` uses Agent, tool, handoff and run
- Reviewer handoffs plus trace flush; fallback results preserve demo reliability

Repo controls and runtime

- AGENTS.md: shared logic in `lib/` ; trust in `auth` / `permissions`
- `chat-extensions.ts` is the MCP / RAG extension point
- OpenNext + Wrangler deploy to Cloudflare Workers; Vitest covers routes and UI